Practical No. 1

Basics of Python Programming

Q1. Write a Python program to display “Hello World”.

```
print("Hello World")
```

Q2. Write a Python program to display your Name, Roll Number, and Class.

```
print("Name: Rahul Sharma")
print("Roll No: 25")
print("Class: FYBSc DS")
```

Q3. Write a Python program to create variables of different data types and display their values and types.

```
name = "Rahul"
age = 18
percentage = 85.5
status = True

print(name, type(name))
print(age, type(age))
print(percentage, type(percentage))
print(status, type(status))
```

Q4. Write a Python program to accept two numbers from the user and perform arithmetic operations.

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

print("Addition =", a+b)
print("Subtraction =", a-b)
print("Multiplication =", a*b)
print("Division =", a/b)
```

Q5. Write a Python program to calculate the area of a rectangle.

```
length = float(input("Enter length: "))
breadth = float(input("Enter breadth: "))

area = length * breadth
```

```
print("Area =", area)
```

Q6. Write a Python program to calculate the area and circumference of a circle.

```
radius = float(input("Enter radius: "))

area = 3.14 * radius * radius
circumference = 2 * 3.14 * radius

print("Area =", area)
print("Circumference =", circumference)
```

Q7. Write a Python program to convert temperature from Celsius to Fahrenheit.

```
celsius = float(input("Enter temperature in Celsius: "))

fahrenheit = (celsius * 9/5) + 32

print("Fahrenheit =", fahrenheit)
```

Q8. Write a Python program to swap two variables.

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

print("Before Swap:", a, b)

a, b = b, a

print("After Swap:", a, b)
```

Q9. Write a Python program to demonstrate type conversion.

```
num = "100"

print(type(num))

num = int(num)

print(type(num))
print(num + 50)
```

Q10. Write a Python program to evaluate arithmetic expressions.

```
a = 10
b = 5
```

```
print(a + b)
print(a * b)
print(a / b)
print(a ** b)
```

Q11. Write a Python program to demonstrate relational operators.

```
a = 10
b = 20

print(a > b)
print(a < b)
print(a == b)
print(a != b)
```

Q12. Write a Python program to demonstrate logical operators.

```
a = 10

print(a > 5 and a < 20)
print(a > 20 or a < 20)
print(not(a > 5))
```

Q13. Write a Python program to calculate simple interest.

```
p = float(input("Enter Principal: "))
r = float(input("Enter Rate: "))
t = float(input("Enter Time: "))

si = (p * r * t) / 100

print("Simple Interest =", si)
```

Q14. Write a Python program to calculate the average of three numbers.

```
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))
c = float(input("Enter third number: "))

avg = (a + b + c) / 3

print("Average =", avg)
```

Q15. Write a Python program to accept a student's name and marks in three subjects and calculate total and percentage.

```
name = input("Enter student name: ")

m1 = float(input("Enter marks of Subject 1: "))
m2 = float(input("Enter marks of Subject 2: "))
```

```
m3 = float(input("Enter marks of Subject 3: "))

total = m1 + m2 + m3
percentage = total / 3

print("Student Name =", name)
print("Total Marks =", total)
print("Percentage =", percentage)
```

Demonstration in Python IDLE

For every program, teach students:

1. Open **Python IDLE**.
2. Click **File → New File**.
3. Type the program.
4. Save the file (`Ctrl + S`).
5. Save with `.py` extension.
6. Press **F5** or **Run → Run Module**.
7. Enter input values (if required).
8. Observe the output in the Shell window.